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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 01/31/2011 |
| **PI Name:** | Hlatky, Lynn Ph.D. | | |
| **Project Title:** | NSCOR: Solid Tumor Risk Estimation: Incorporating Intercellular Interaction Effects | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02135-2907 | **Congressional District:** | 8 |
| **Comments:** | PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2004 NSCOR Space Radiation NNH04ZUU002N |
| **Start Date:** | 10/01/2004 | **End Date:** | 10/31/2010 |
| **No. of Post Docs:** | 10 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 12 |
| **No. of Bachelor's Candidates:** | 12 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Cucinott1a, Francis | **Contact Phone:** | 281-483-0968 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date should be 10/31/2010, per M. Klumpar in PI's office [Ed., 3/28/2011]<br>NOTE: Received NCE to 3/31/2011 per C. Guidry/JSC (10/2010)<br>NOTE: Received NCE to 9/30/2010 per J. Dardano/JSC (9/09)<br>NOTE: New grant number NNJ06HA28G received in 2006 when PI moved to Tufts and end date is 9/30/2009 (information received 6/2009) jvp<br>NOTE: End date changed back to 9/30/2009 per S. Krenek/JSC (8/07)<br>End date changed to 10/31/2009 per JSC info update (10/06) | | |
| **Key Personnel Changes/Previous PI:** | Dr J. Folkman deceased | | |

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| **COI Name (Institution):** | Folkman, Judah M.D. ( Children's Hospital, Harvard Medical School )<br>Hahnfeldt, Philip Ph.D. ( Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine )<br>Huber, Peter ( German Cancer Research Center (DKFZ) )<br>Lamont, Clare ( Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine )<br>Sachs, Rainer Ph.D. ( University of California )<br>Abdollahi, Amir ( Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine )<br>Nickerson, Jeffrey ( Univ. of Massachusetts Medical School ) |
| **Grant/Contract No.:** | NNJ06HA28G ; NNJ04HJ12G |
| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | We will improve solid tumor risk estimation for astronauts. Improvements will be based in large part on analyzing data resulting from our high-energy iron ion and proton in vivo irradiations. Our mathematical risk modeling will use standard computational approaches which minimize theoretical assumptions and also use minimally-parameterized, biologically-based carcinogenesis models capable of interrelating epidemiological data, animal experiments, and in vitro radiobiology. In our experiments and our computational analyses, special emphasis will be placed on tumor promotion and progression, as influenced by intercellular signaling among various cell populations within the tumor (e.g. tumor cells, endothelial cells and stroma).<br>The in vivo models used are highly suitable for the program. These include the transgenic conditional k-ras lung tumor model from the Tyler Jacks's laboratory that will allow the study of radiation modulation of all steps in carcinogenesis. These mice have a recombinant adenovirus expressing Cre recombinase to induce k-ras expression which synchronizes tumor onset, allowing numerous tumors to develop simultaneously in the lungs. The number of tumors and rate as which they progress though the four stages of carcinogenesis (e.g. time-dependent ratio of hyperplastic to neoplastic lesions) can then be traced as a function of the timing, dose and quality of the radiation. We will also use dormant human breast tumor xenograft models which allow us to place particular emphasis on dissecting the progression stage of carcinogenesis. These in vivo models offer specific advantages: solid tumor sites of concern to NASA (lung, breast, and later thyroid) will be studied; middle-age mice will be used; and due to the nature of these models comparatively few mice will be needed and post-irradiation study times will be comparatively short.<br><br>In this Program we implement a multi-level approach with whole organism-, tissue- cell-, and molecular-level endpoints used to measure radiation response. For examples, in addition to radiation influence on latency periods or the time-dependent ratio of dysplastic versus frankly neoplastic lung lesions, we examine cell signaling, use matrix and clustering computer algorithms for analyzing transcriptome data, and interpret chromosome aberrations scored with mFISH/SKY using our established computer simulation software.<br><br>Importantly for our emphasis on intercellular signaling as a key aspect of carcinogenesis, we will assay not only tumor cells but also tumor-associated stromal and endothelial cells (including circulating endothelial cells). It is turning out that such tissues can help support or repress tumors (i.e. play a major role in promotion and progression), and due to the relative stability of their genomes they are simpler to analyze for radiation-induced DNA damage. An experimental and theoretical emphasis on tumor progression is also planned, based on the fact that this step in carcinogenesis has hitherto received less attention from radiation risk modelers than other steps, e.g. initiation, but is at least as important. There is now strong evidence that microscopic dormant neoplastic sites are far more prevalent in adults than previously assumed and that their progression can be accelerated by radiation. Radiation shortening of latency periods could thus be a key component of solid tumor risk for middle-aged astronauts.<br><br>A tightly-knit interdisciplinary team, Director L. Hlatky, Associate Director R. Sachs along with the other Project leaders J. Folkman, P. Huber, and P. Hahnfeldt will carry out the 5 closely interrelated Projects: (1) mouse models for assessing carcinogenesis risk; (2) HZE and low LET irradiation; (3) radiation transcriptome analyses; (4) quantitative chromosome aberration analysis; and (5) quantitative estimation of solid tumor risk. Project (5) will integrate data from the first four Projects as well as drawing on results from the literature. It will culminate in a composite radiation carcinogenesis model designed to reduce the uncertainties in risk estimates for astronauts by addressing cell interaction effects and including in the multi-step carcinogenesis analysis the overlooked step of progression. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Autopsy findings from adults dying from non-cancer causes have now unequivocally established that most of us possess fully-malignant tumors that are held in check by a balance between cell cycling and cell death. Such tumors are characterized by a failure to advance through the angiogenic switch. Dormant tumors that manage to switch to the angiogenic phenotype grow exponentially and ultimately become clinically apparent, which is referred to as the "disease state" of cancer (Folkman, Nature 2004).<br>As recent results from Project 2 disclose, major angiogenic genes show significant regulation by space radiations. This suggests there may be a link between space radiation exposure and modulation of the dormancy bottleneck that holds existing indolent lesions in check. To research the matter, the development of dormant tumor models has been a priority. We now have in place several human dormancy models in mice. The models have been phenotypically characterized (Almog and Folkman, JNCI 2006 and FASEB 2004). Tumors remain dormant for a prolonged period of time until they spontaneously switch to the fast-growing phenotype. The time point of the switch is tumor-type dependent. We recovered several of the spontaneously-switched tumors from mice and cultured them in vitro. Re-implantation in naive mice confirmed a stable fast-growing phenotype. It is concluded a genetic reprogramming event accompanies transition from dormancy to fast-growth.<br><br>We have designed new methods for estimating the incidence of iatric (i.e. treatment induced) cancers after radiotherapy. As cancer radiotherapy methods improve and the number of patients treated increases, more and more people survive long enough after the initial cancer diagnosis to potentially develop radiotherapy-induced second malignancies. Because treatment techniques are changing rapidly, there is insufficient follow-up time to directly measure the second cancer risks associated with the newer methods. Biologically-based mathematical models, which implement plausible assumptions about radiation carcinogenesis processes such as initiation of organ-specific stem cells to the pre-malignant state, inactivation (killing), and repopulation of both normal and pre-malignant stem cells during radiotherapy and subsequent organ recovery, can address this problem: Models calibrated with older data can predict second cancer risks |

of new/prospective radiotherapy protocols.

**Task Progress:**

Overview.
At the time we proposed this NSCOR, the idea that intercellular interactions may impact radiogenic cancer risk at multiple scales was not well-appreciated, nor was the notion that modulation of cancer latency and progression can alter cancer risk. Taking a practical view of cancer risk as that of eventual disease 'awareness' and clinical presentation, rather than cancer cell creation, we described how many factors alter risk by derailing or expediting the long course from cell initiation to manifest disease. To uncover these factors and their consequences, we used a combination of mathematical, molecular biological and functional assays in combination with animal models to study the full carcinogenesis timeline. Strikingly, not only were we able to confirm our hypothesis that extracellular, population-level influences can impact risk, but that cancer risks from proton and Fe radiations may differ in large part due to their effects on the tumor-host interaction. Angiogenesis gene regulations and host age figured prominently in the distinction. Theoretically, we showed that the tumor as an interactive population of cells takes on emergent dynamic qualities beyond those of its constituent cells. Risk models generated from these considerations offer alternative means to explain epidemiologic risk data. Finally, a radically new cell-cell interaction mechanism was uncovered that not only changes how radiogenic cancer risk should be assessed, but recasts our basic understanding of carcinogenesis itself.

Background.

Standard radiobiology methods for estimating solid tumor risk do not yet give sufficient accuracy for space flight planning. RBEs for the mixture of radiations encountered in space are one major source of uncertainty. Also poorly understood are radiation effects on tumor promotion and progression. Beyond initiation and transformation, the promotion and progression steps, where initiated or transformed cell lineages proliferate and evolve, are turning out to be critically important control stages in carcinogenesis. Promotion and progression are population-level processes, strongly influenced by microenvironment and intercellular, tissue-level (e.g. tumor-stromal and angiogenic tumor-endothelial) signaling. Such intercellular interactions between different cell types are critical to the maintenance of tumor latency; perturbing intercellular interactions can lead to accelerated growth of neoplasms. Cancer presentation due to the radiation-induced advancement of any pre-existing neoplasm will likely occur ahead of any cancer presentation due to a radiation-initiated cell. Consequently, the threat posed by disrupting latency of a pre-existing neoplasm is particularly significant in space travel. Moreover, strong evidence points to a far greater prevalence of pre-existing, undetected microscopic neoplasms in adults than was previously thought. Thus, the consideration of radiation influences on tumor promotion, and especially on tumor progression, is very important for the assessment of solid cancer risks in (middle-aged) astronauts.

Evidence for tissue-level control of neoplastic development is abundant. A case in point is the achievement of oncogenic potential of normal epithelium in the forestomach and prostate when their associated fibroblasts lose TGF-beta receptor II signaling. Diminished expression of the cyclin-dependent kinase inhibitors p21 and p27 (tumor-suppressor genes) as well as increased expression of the c-Myc oncogene are both facilitated by this stromal alteration. Implied is a description of carcinogenesis as involving irreversible progressive genetic dysfunction in cells, the expression of which is under the control of tissue communication between the prospective tumor cells and associated stroma. We proposed that the influence of tissue communication on functional cancer risk, i.e., the likelihood of disease presentation, may on occasion be decisive. Remarkably, it turns out essentially all adults already have dormant microscopic neoplasms that will not present in their lifetimes. In an autopsy study of people who were thought to be cancer-free at the time of death, the frequency of microscopic in situ neoplastic lesions detected by histological examination was found to be substantial. As it turns out, tumors of the thyroid and breast, considered high on a list of types most at risk for radiation induction, also stood out as pervasive but undiagnosed lesions at autopsy in the Black and Welch study. This suggests that the increased incidence of these types of cancer following radiation exposure may well be due to progression of such dormant in-situ lesions.

Guiding our work, therefore, was the overarching hypothesis that "inappropriate signalings between the epithelium and its associated tumor-supportive compartments (vasculature and stroma) play promoting and permissive roles in the carcinogenesis program for the prospective tumor cell. It is therefore important for the proper assessment of risk that dose and radiation quality effects on these communications be elucidated and quantified."

Specific Objectives.

Using animal models developed under Project 1, we explored how final cancer risk is influenced by environmental and population factors modulating tumor development, with emphasis on how specific controls governing tumor dormancy in vivo may also forestall or even eliminate radiogenic cancer risk. The aim of Project 2 was to use animal and functional assays to compare proton and Fe irradiations to gamma in their ability to transform cells and to progress existing tumors. Project 3 asked these questions at the molecular biological level. Projects 4 and 5 considered the theoretical interpretations, both at the level of the developing tumor cells as a population, and as a population interacting with the host. The objective of these projects was to explain the empirical findings of the other projects in terms of a unifying quantitative mechanism that could then be used to inform final disease risk.

Significance of the Work.

The significance of the work lies in the challenge it presents to the standard radiogenic cancer paradigm. This paradigm holds that cancer may be understood as an autonomous expansion of an initiated cell clone, followed by the chance genetic creation of a cancer cell in that clone that then inevitably expands to become a diagnosed cancer. At the basis of these events is thought to be the DNA lesion. We argue that essential population-level and tumor-environmental dynamics can alter this picture dramatically, either promoting cancer development or slowing the course of existing cancers. Their basis extends beyond DNA damage to include the familiar "hallmarks" of cancer that constitute "bottlenecks" to tumor advancement – population-level logistic growth effects, acquisition of angiogenic potential, and competency for invasion, to name three. We have shown that radiation can progress or limit cancers by modifying these bottlenecks. Because these bottlenecks may be significant enough to increase the likelihood of transforming an initiated cell, or to change the likelihood an existing cancer advances to clinical detection over a lifetime, they must be included in our overall assessment of radiogenic cancer risk. It follows that existing models which fail to include such bottleneck effects are inherently flawed in that they do not include all radiation actions bearing on the development of manifest cancer disease.

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